

xargs Cheat Sheet

Turn a list of words or lines into arguments for another command.

Why it exists

Many commands (**rm**, **cp**, **chmod**, **kill**) ignore piped input; they only act on names given as **arguments**. **xargs** bridges the gap: it reads input, then builds and runs the command with that input on the end. Huge lists are split into several runs, avoiding *Argument list too long*.

```
find . -name '*.tmp' | rm    ← does nothing useful: rm doesn't read the pipe
find . -name '*.tmp' | xargs rm ← runs: rm ./a.tmp ./b.tmp ./c.tmp
```

Using it

```
command | xargs [options] cmd [fixed args] ← input items are added to the end of cmd
command | xargs -I {} cmd ... {} ...       ← input goes wherever {} appears instead
```

Option	What it does	Option	What it does
-n 1	Uses at most 1 item per command run	-0	Items end in a null byte (for -print0)
-I {}	Replaces {} with the item; one run per line	-d '\n'	Splits on new lines only (GNU)
-L 1	Uses 1 input line per command run	-P 4	Runs up to 4 commands at the same time
-t	Prints each command before running it	-r	Doesn't run at all if input is empty (GNU)
-p	Asks y/n before each command	-a file	Reads items from a file, not the pipe

How input is split

By default **xargs** splits at **any space, tab or new line**, and treats quotes and backslashes specially. So a file called **my file.txt** becomes two items, **my** and **file.txt**. Fixes:

Safe pattern	Why
find . -name '*.log' -print0 xargs -0 rm	Null bytes can't appear in file names, so any name works (best)
... xargs -d '\n' rm	One item per line; spaces are kept (GNU only)
... xargs -I {} rm {}	One item per line, but runs rm once per file (slower)

Reading the output

```
$ printf 'a\nb\nc\n' | xargs -t -n 2 echo
echo a b    ← -t: the command xargs is about to run (printed to stderr)
a b         ← that command's own output
echo c      ← -n 2 means 2 items per run, so a second run picks up the rest
c
```

Exit status: **0** = every run succeeded, **123** = some run failed, **127** = command not found. With **-P**, output from runs can be mixed.

Handy recipes

Command	Does
find . -name '*.tmp' -print0 xargs -0 rm -f	Deletes temp files safely, even with spaces in names
grep -rl 'old' . xargs sed -i 's/old/new/g'	Find-and-replace in every matching file (GNU sed)
ls *.jpg xargs -I {} cp {} /backup/{}.bak	Copies each file with a new name
cat hosts.txt xargs -n 1 -P 8 ping -c 1	Pings 8 hosts at a time
pgrep -f myjob xargs -r kill	Kills matching processes; does nothing if none
ls *.gz xargs -n 1 -P 4 gunzip	Unpacks files 4 at a time
echo a b c d xargs -n 2	Regroups words, 2 per line (default command: echo)

Remember: for file names, always use **find ... -print0 | xargs -0**. Not sure what it will run? Put **echo** in front of the command, or add **-t** / **-p**, and check first.